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***B.Tech. Degree II Semester Regular and Supplementary/
I Semester Supplementary Examination May 2017***

**CS/EC/IT AS 15-1202 B & CE/EE/ME/SE AS-15-1102 A
ENGINEERING PHYSICS
(2015 Scheme)**

Time : 3 Hours

Maximum Marks : 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Write a brief note on the properties of lasers.
 (b) Discuss the principle of Holography with a diagram.
 (c) What are the different types of dispersion in optical fibers?
 (d) What do you mean by packing factor? Calculate packing factor for simple cubic, BCC and FCC structures.
 (e) Name the different types of liquid crystals. Describe their features.
 (f) What do you mean by pseudo elasticity in shape memory alloys?
 (g) Write a brief note on the applications of nanomaterials.
 (h) What are type-I and type-II superconductors.
 (i) What do you mean by a black body? Write the Plank's formula for black body radiation.
 (j) What is reverberation? Describe how the reverberation time affects the sound level in a hall.

PART B

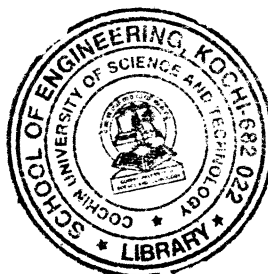
(4 × 10 = 40)

- II. (a) What do you mean by Fermi energy level in a semiconductor? How does the position of Fermi level affect the working of semiconductor laser? Discuss the construction and working of semiconductor laser. (8)
 (b) A pulsed laser is constructed with a ruby crystal as the active material. The ruby rod contains typically a total of 3×10^{19} Cr^{3+} ions. If the laser emits light at 6943 Å wavelength, find (i) the energy of emitted photon in eV and (ii) the total energy available per laser pulse (assume total population inversion) (2)

OR

- III. (a) What are optical fibres? Discuss different types of fibers with appropriate diagrams and applications. (8)
 (b) The velocity of light in the core of a silica fiber is 2×10^8 m/s and the critical angle of core cladding interface is 60° . Calculate (i) the refractive indices of core and cladding and (ii) the numerical aperture of the fiber. (2)

(P.T.O.)



- IV. (a) Derive Bragg's law. Discuss the crystal structure analysis using Bragg's X-ray spectrometer. (7)
 (b) Lead is face centered cubic with atomic radius of $1.746 \text{ \AA}$. Find the spacing of (i) (200) planes, and (ii) (220) planes. (3)

OR

- V. (a) Describe the construction and working of liquid crystal display system with a neat diagram. What are the merits and demerits of the LCD? (7)
 (b) What are metallic glasses? Write a note on the properties of metallic glasses. (3)

- VI. (a) What are superconductors? Discuss the following terms (i) Meissner effect (ii) Critical current density (iii) Isotope effect (iv) high temperature superconductors. (8)
 (b) Discuss the classification of nanomaterials. (2)

OR

- VII. (a) Derive Einstein's mass energy relationship. (8)
 (b) The density of gold is $19.3 \times 10^3 \text{ kg/m}^3$ in a frame S which is at rest. Calculate its density for an observer in a frame S', which is moving along x-axis with a speed of $0.9c$. (2)

- VIII. (a) Derive time independent Schrodinger equation. What is the significance of wave function in quantum mechanics? (7)
 (b) State and explain Heisenberg's uncertainty principle. Calculate the uncertainty in the position of an electron moving with a speed of 300 m/s and if the uncertainty in its velocity is 0.001% . (3)

OR

- IX. (a) Describe the principle, construction and working of ultrasonic generator based on piezoelectric effect. (7)
 (b) What is the reverberation time of a cubic hall if a false ceiling is made at half the height of the hall (at 7 feet). The absorption coefficient of the false ceiling materials is 0.3 and that of wall and floor is 0.25 . Analyzing your result, suggest suitable actions needed (if any) to maintain acoustic efficiency. ($1 \text{ foot} = 0.3048 \text{ m}$) (3)
